CS336 Assignment 3 (scaling): Scaling Laws

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Spring 2026

1 IsoFLOPs Scaling Laws (`chinchilla_isoflops`)

(a) Optimal Model Size

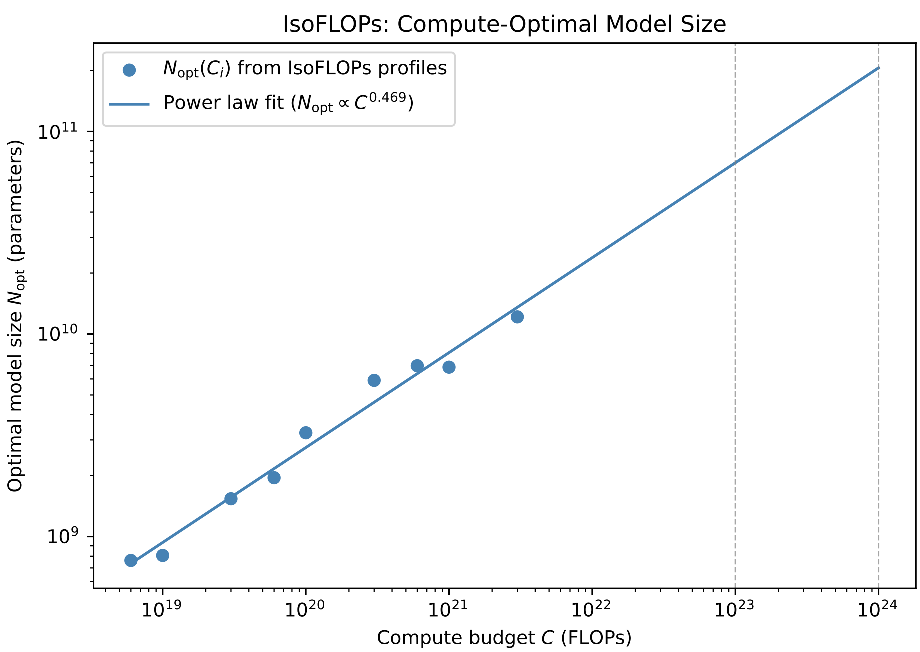

Figure 1: Best model size N_{opt} for a fixed compute budget C. Points come from IsoFLOPs profiles. The line is a power fit $N_{\mathrm{opt}} \propto C^a$.

At $C = 10^{23}$ FLOPs the best model size is 7.0×10^{10} parameters. At $C = 10^{24}$ FLOPs it is 2.1×10^{11} parameters.

(b) Optimal Dataset Size

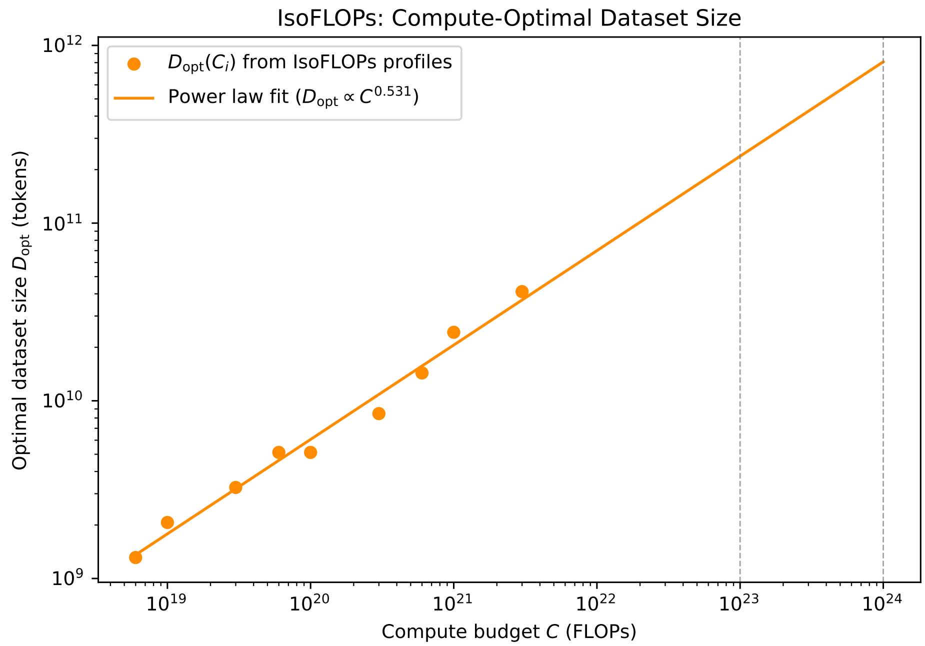

Figure 2: Best dataset size D_{opt} for a fixed compute budget C. Points come from IsoFLOPs profiles. The line is a power fit $D_{\text{opt}} \propto C^b$.

At $C = 10^{23}$ FLOPs the best dataset size is 2.4×10^{11} tokens. At $C = 10^{24}$ FLOPs it is 8.1×10^{11} tokens.

2 Constructing Scaling Laws (`scaling_laws`)

2.1 Experimental Design

We split a 12-hour B200 budget into three phases. The plan follows Kaplan et al. [1], Hoffmann et al. [2], and Yang et al. [3]. First we ran one short job. It checked the API, the logging path, and how much of peak FLOPs the cluster reached in practice (MFU).

Phase 1: learning rate on a small model. We need one learning rate for all later runs. We tuned it on the smallest model allowed by the API: `hidden_size` = `num_attention_heads`·`head_dim` with `head_dim` = 64. That model has $d_{\text{model}} = 128$, $L = 2$, and 1.77 M non-embedding parameters. We tried nine learning rates on a log grid from 3×10^{-4} to 3×10^{-1}. Each run used about 16 M tokens and a cosine schedule over that length (Hoffmann et al. [2], their Approach 3). The upper rate was set high on purpose so the largest value would fail. That way the best rate sits inside the grid, not on the edge.

Phase 2: how learning rate scales with width. Yang et al. [3] describe μP, where the best learning rate does not depend on width. The full recipe needs options the assignment API does not expose. We used the simpler width-scaling rule from Yang et al. [3] (Section 6) and Hoffmann et al. [2]:

$$\eta(d) = \eta_{\text{proxy}} \cdot \left(\frac{d_{\text{proxy}}}{d}\right)^{\gamma}. \tag{1}$$

We treat γ as a number to fit from data. We tried three learning rates at $d_{\text{model}} \in \{320, 512\}$ around the common guess $\gamma = 1.0$. The first grids put the best rate at the low end ($\times 0.5/\times 1/\times 2$). We added a second round at $\times 0.25$ and $\times 0.5$ of the previous best. After that, the best rate lay inside the grid for both widths.

Phase 3: scaling-law data. We fixed η_{proxy} and γ. We trained six model families from 1.77 M to 116 M non-embedding parameters. Compute targets were $C \in \{10^{18}, 3 \times 10^{18}, 10^{19}\}$ FLOPs (Table 1). We kept depth and width near $d_{\text{model}}/L \approx 50$. Kaplan et al. [1] show that shape matters little at fixed parameter count. Two compute levels (10^{18} and 3×10^{18}) have at least four families each. That lets us compare IsoFLOPs points to the parametric fit (Hoffmann et al. [2] Approach 2).

(d_{model}, L)	N (non-embedding)	Compute levels (FLOPs)
$(192, 4)$	1.77 M	$\{1\times10^{18}\}$
$(320, 6)$	7.37 M	$\{1\times10^{18},\ 3\times10^{18}\}$
$(448, 9)$	21.7 M	$\{1\times10^{18},\ 3\times10^{18},\ 1\times10^{19}\}$
$(576, 10)$	39.8 M	$\{1\times10^{18},\ 3\times10^{18},\ 1\times10^{19}\}$
$(704, 12)$	71.5 M	$\{3\times10^{18},\ 1\times10^{19}\}$
$(832, 14)$	116.4 M	$\{1\times10^{18},\ 3\times10^{18}\}$

Table 1: Phase 3 model families and per-family compute levels.

Budget handling. The API reserves wall time up to `max_runtime_seconds` when a job starts. It returns unused time when the job ends. We set `max_runtime` to `safety_factor` $\cdot\, C/($`MFU` $\cdot$ `peak_FLOPs`) with `MFU` = 0.4 and `safety_factor` = 1.5. Values above the two-hour cap were clipped. Phase 3 went out in three batches (five, five, three runs). Earlier batches freed budget before the next batch started.

2.2 Phase 1 & 2 Results

The small-model sweep peaked at $\eta_{\text{proxy}} = 2.25\times10^{-2}$ (validation loss 5.85). The endpoints 3×10^{-4} and 3×10^{-1} were worse (7.98 and 6.68). So the grid surrounds the best value.

Phase 2 gave best rates 4.50×10^{-3} at $d_{\text{model}} = 320$ and 2.81×10^{-3} at $d_{\text{model}} = 512$. Plugging into the scaling law and averaging over the two widths,

$$\gamma \;=\; \frac{1}{2} \sum_{d\in\{320,\,512\}} \frac{\log\big(\eta(d)/\eta_{\text{proxy}}\big)}{\log(d_{\text{proxy}}/d)} \;=\; 1.628. \tag{2}$$

This is larger than the often-used $\gamma = 1$. Yang et al. [3] (Figure 1) show the simple scaling rule can shift the best rate by almost an order of magnitude when width grows. We use $\gamma = 1.628$ in Phase 3.

2.3 Phase 3 Outcome and Anchor Calibration

All 13 Phase 3 runs hit the time cap. From eval throughput we estimated real MFU between 1.6% and 4.7%. We had assumed 40% for planning. Three smallest jobs logged no eval block. The other ten gave one to three partial validation losses each.

We still used the partial runs. For each we set trained tokens to $D_{\text{actual}} = (\texttt{evals\_completed}/16) \cdot D_{\text{planned}}$. We fit the law on $(N, D_{\text{actual}}, L_{\text{partial}})$. Stopped runs may bias loss upward: the schedule was cut while the learning rate was still high.

Anchor run. We checked that on one setup: $(d_{\text{model}}, L) = (576, 10)$. We trained again with `total_train_tokens` equal to D_{actual} from the truncated run. The cosine run could finish as intended (final LR $0.1\,\eta_{\text{peak}}$). The two losses match within 0.001 nats:

- Truncated run (high LR throughout): $L = 3.7820$.
- Full schedule on D_{actual}: $L = 3.7813$.

The effect is tiny at this (N, D). We fit using salvaged D_{actual} values.

2.4 Fitting the Scaling Law

We used Hoffmann et al. [2] Approach 3 (their Equations 2–3). The model is

$$\widehat{L}(N, D) \;=\; E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}. \tag{3}$$

We estimated (E, A, B, α, β) with Huber loss on log-residuals ($\delta = 10^{-3}$, as in Hoffmann et al.) and L-BFGS-B. We started from eight initial guesses: $\log E \in \{\log 1, \log 2\}$, $\alpha \in \{0.3, 0.5\}$, $\beta \in \{0.3, 0.5\}$. We kept the run with the lowest objective.

On the ten salvaged points,

$$E = 3.376, \quad A = 9.66 \times 10^{6}, \quad B = 9.64 \times 10^{6}, \quad \alpha = 1.049, \quad \beta = 0.857.$$

Our α and β are larger than Hoffmann et al. [2]'s ($\alpha = 0.34$, $\beta = 0.28$). The ratio $\alpha/\beta = 1.22$ is almost the same as Chinchilla's 1.21. The best split between N and D at fixed compute depends only on that ratio (Hoffmann et al. [2] Equation 5). So our allocation prediction is in line with theirs. The larger exponents likely come from our narrow N range (5.4× here vs. 230× in Hoffmann et al.).

2.5 Quality of Fit

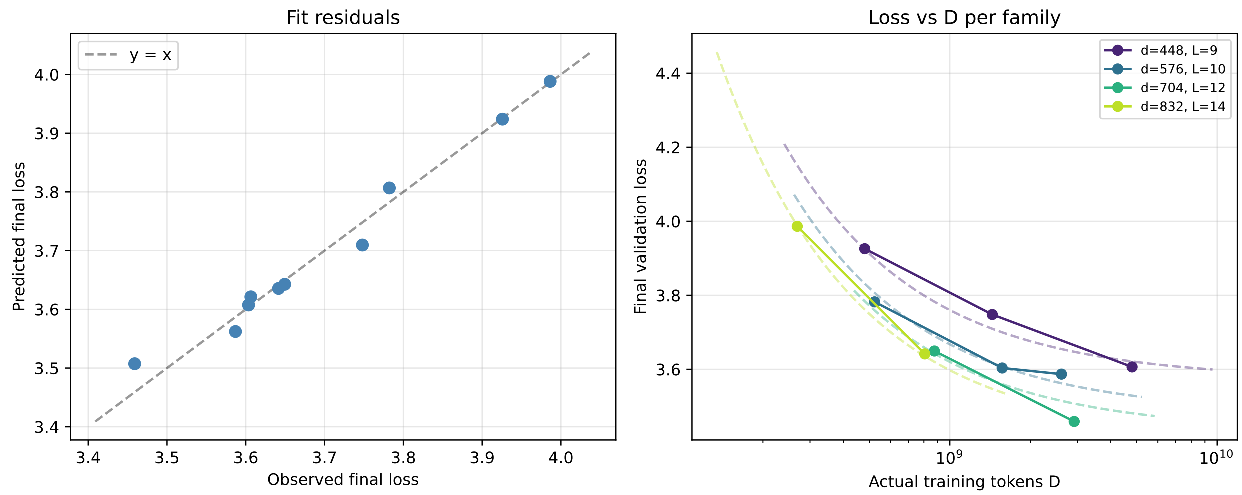

Figure 3: Left: predicted vs. observed loss for the ten salvaged Phase 3 runs. Right: loss vs. tokens by model family; dashed lines are the fit at fixed N.

All ten residuals are small ($\max|r| = 0.048$ nats, RMSE 0.025 nats); see Figure 3. Two limits apply outside this range:

- N only spans 21.7 M to 116 M (5.4×). Hoffmann et al. [2] stress that exponents depend strongly on the range.
- Token counts give $D/N \in [12, 220]$. Most points are above the Chinchilla rule $D/N \approx 20$. Extrapolation to other D/N may be weak.

2.6 Predicted Optimal Configuration for 48 B200-Hours

Hoffmann et al. [2] Equation 5 with unconstrained compute gives $N^* = 1.23 \times 10^9$ and $D^* = 1.05 \times 10^{11}$ for 7.78×10^{20} FLOPs (48 B200-hours at peak). At measured MFU ≈ 5% on our runs, that job would need about 211 wall-clock hours. It does not fit the 48-hour limit.

We minimized $\widehat{L}(N, D)$ under $6ND = M \cdot \texttt{peak\_FLOPs} \cdot 48\,\text{h}$. We set $M = 0.05$ from the 116 M-parameter runs. Larger models often get higher MFU on this hardware. We have no direct measure. We also reduced D by 20% vs. the constrained optimum to keep time slack (≈ 9.6 hours at $M = 0.05$).

Table 2 lists the result. Predicted validation loss:

$$L^* \;=\; \widehat{L}(3.13 \times 10^8,\, 1.66 \times 10^{10}) \;=\; 3.4051. \tag{4}$$

2.7 Hyperparameter Choices

Depth and width follow Kaplan et al. [1]. Schedule and optimizer follow Hoffmann et al. [2]: cosine to total tokens, 10× LR decay, AdamW with $\beta_2 = 0.95$. Peak learning rate follows Yang et al. [3] via the width rule with our $\gamma = 1.628$. That yields $\eta = 6.90 \times 10^{-4}$ from $\eta_{\text{proxy}} = 2.25 \times 10^{-2}$ at $d_{\text{proxy}} = 128$.

Hyperparameter	Value
`num_hidden_layers`	22
`hidden_size`	1088
`num_attention_heads`	17
`head_dim`	64
`intermediate_size`	3072
`total_train_tokens`	1.66×10^{10}
`train_batch_size`	128
Peak learning rate η	6.90×10^{-4}
Warmup fraction	0.05
Final-LR fraction	0.10 (cosine to $0.1\,\eta$)
AdamW (β_1, β_2)	(0.9, 0.95)
Weight decay	0.01
Gradient clip norm	1.0

Table 2: Suggested hyperparameters for the 48 B200-hour run. Non-embedding parameters $N = 3.13 \times 10^8$. The model satisfies `hidden_size` = `num_attention_heads` · `head_dim` and `num_key_value_heads` = `num_attention_heads`.

References

[1] J. Kaplan *et al.*, "Scaling Laws for Neural Language Models." 2020.

[2] J. Hoffmann *et al.*, "Training Compute-Optimal Large Language Models." 2022.

[3] G. Yang *et al.*, "Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer." 2022.